

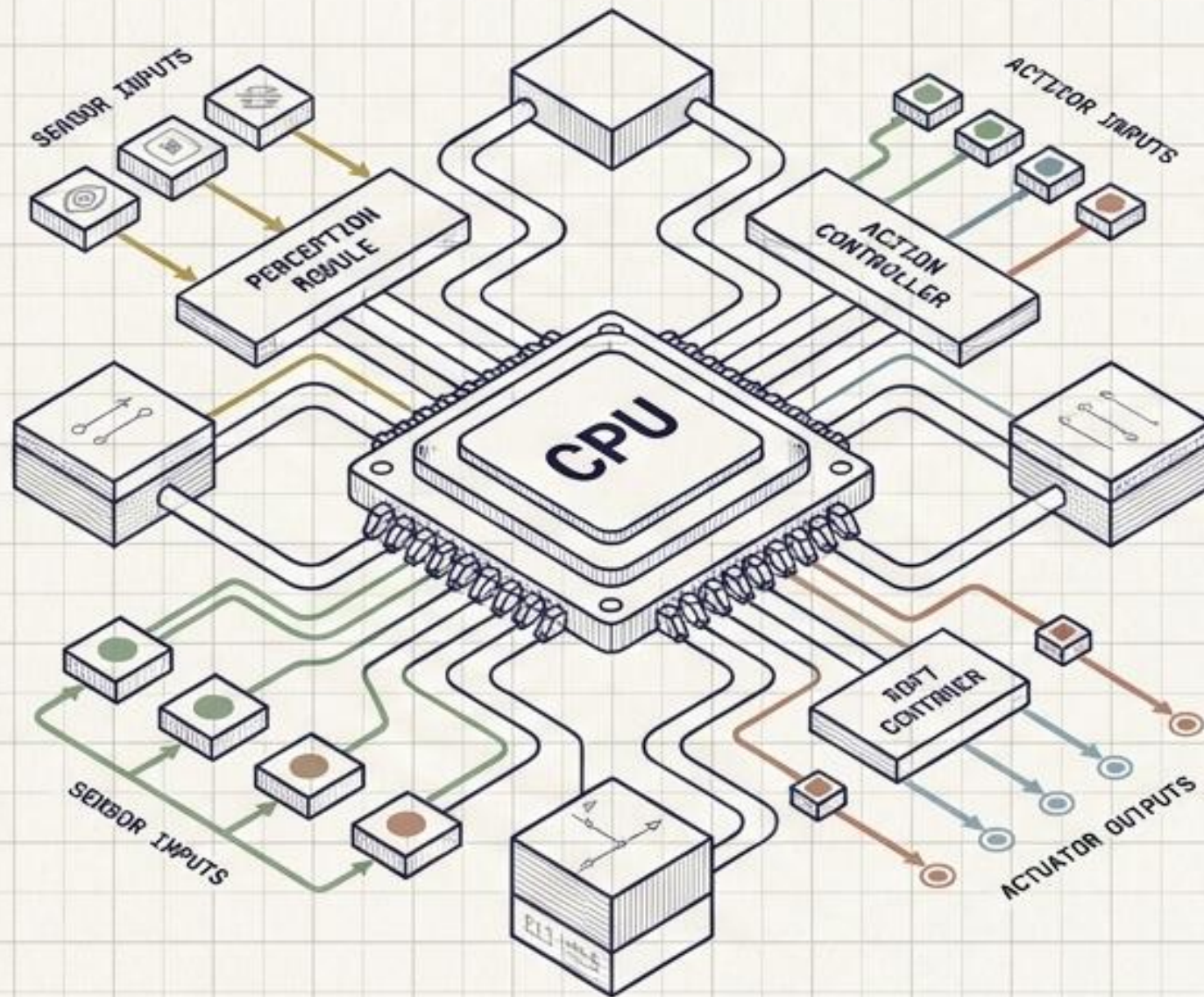


CSC205 ARTIFICIAL INTELLIGENCE | Lecture #5

PEAS: The Architecture of Intelligent Agents

Designing systems that perceive, act, and succeed.

Arifa Awan



Learning Outcomes

By the end of the lecture, students will be able to:

1.



Define and explain PEAS and its role in intelligent-agent design.

2.



Identify Performance Measure, Environment, Actuators, and Sensors for different AI systems.

3.



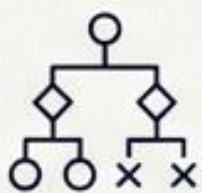
Formulate PEAS descriptions for real-world intelligent agents.

4.



Analyze how PEAS influences agent architecture and decision-making.

5.



Distinguish different task-environment properties such as observable/partially observable, deterministic/stochastic, and static/dynamic.

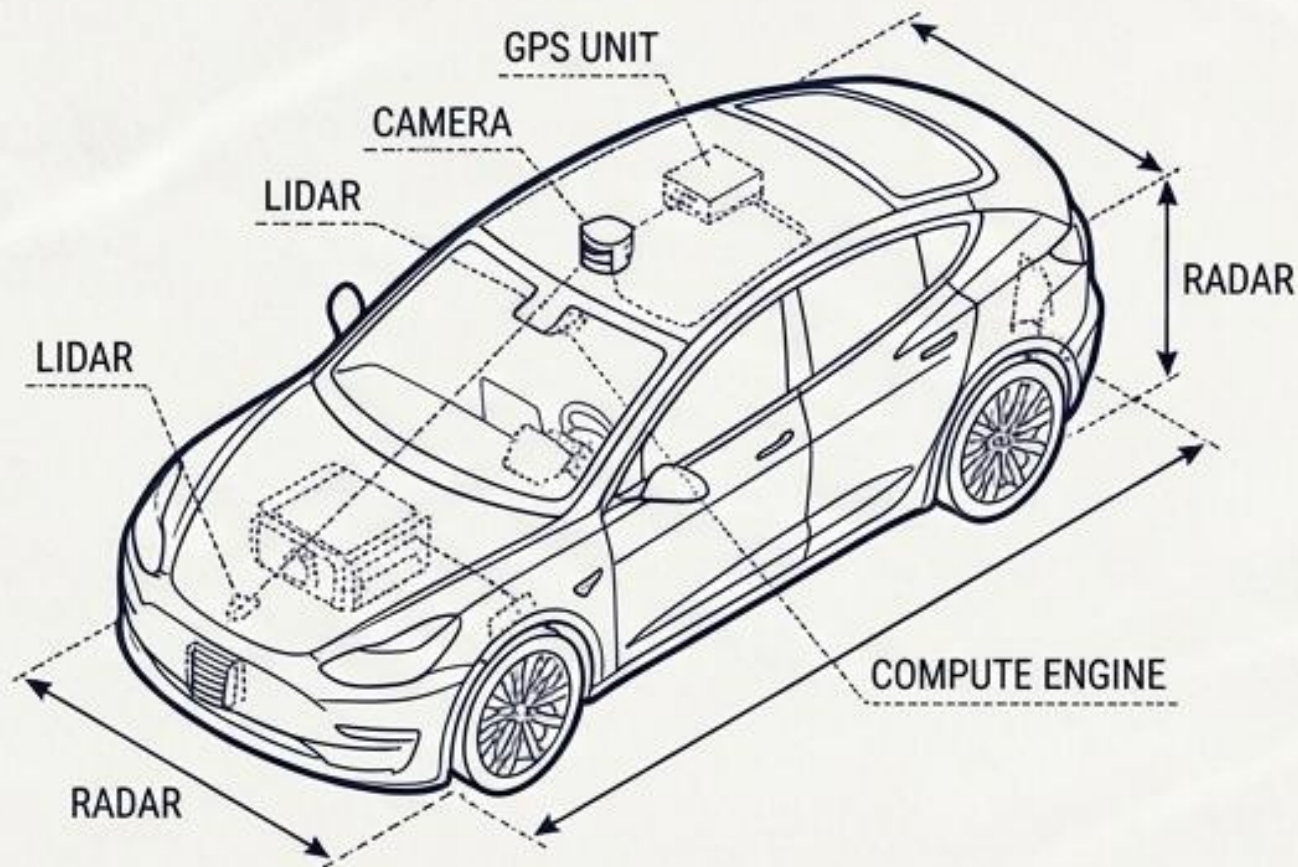
6.



Critically evaluate whether a proposed PEAS specification is complete and appropriate.

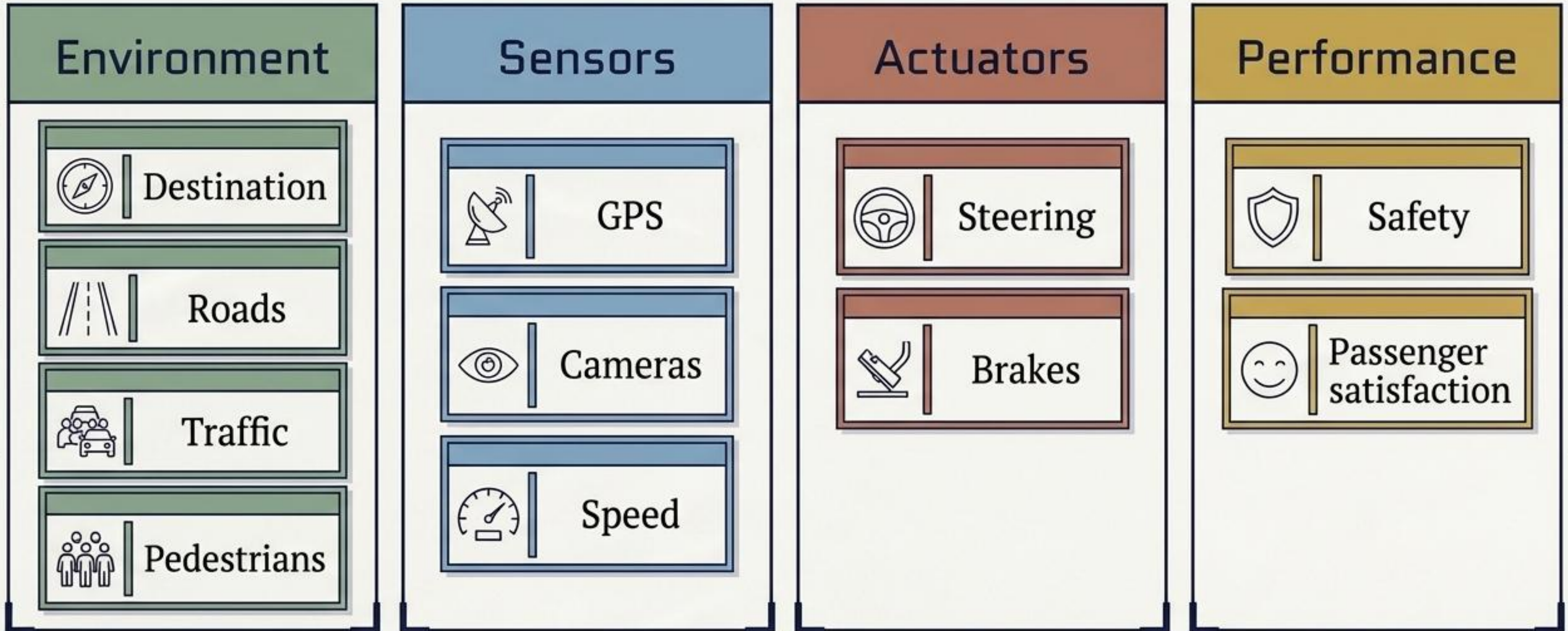
Pre-Assessment: Building an Autonomous Taxi

What information would you need before designing the AI?



Are these all the same type of information? (No. They represent vastly different aspects of the AI agent's task.)

Organizing the Variables

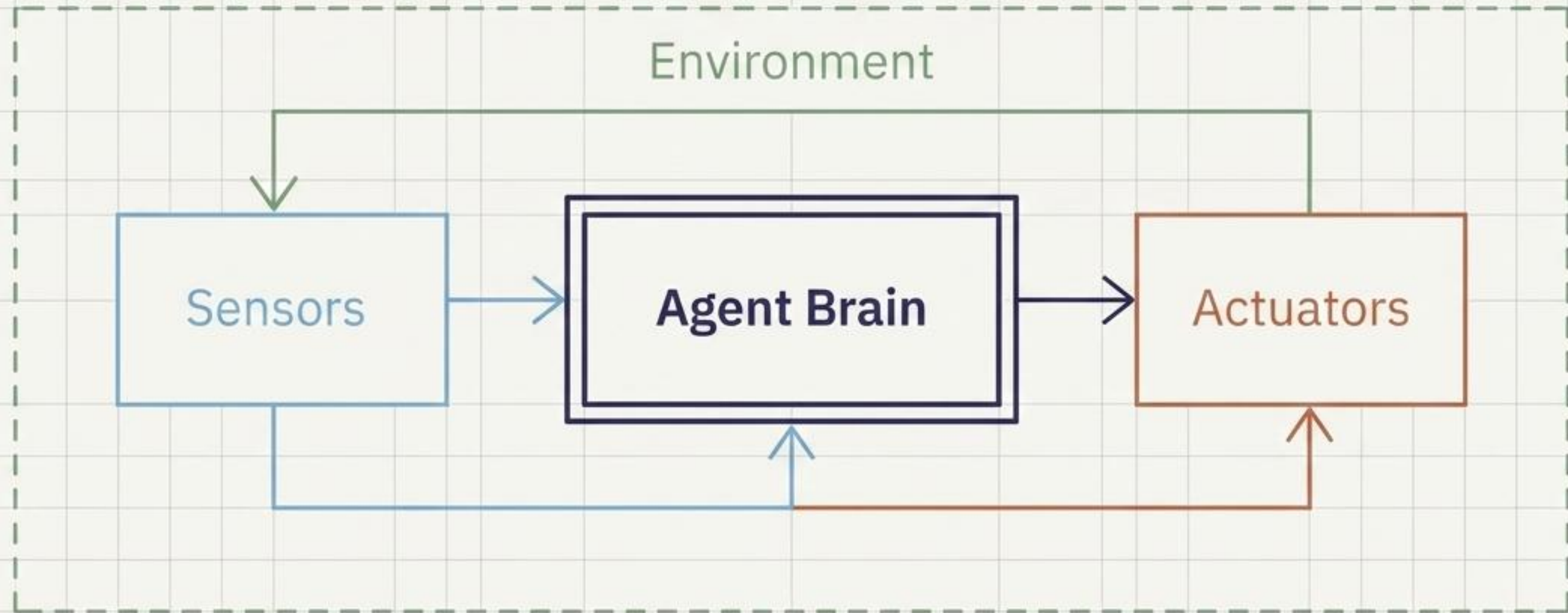


An AI agent cannot be properly designed until we define:

Success Criteria, Operating Environment, Action Capabilities, and Perception Methods.

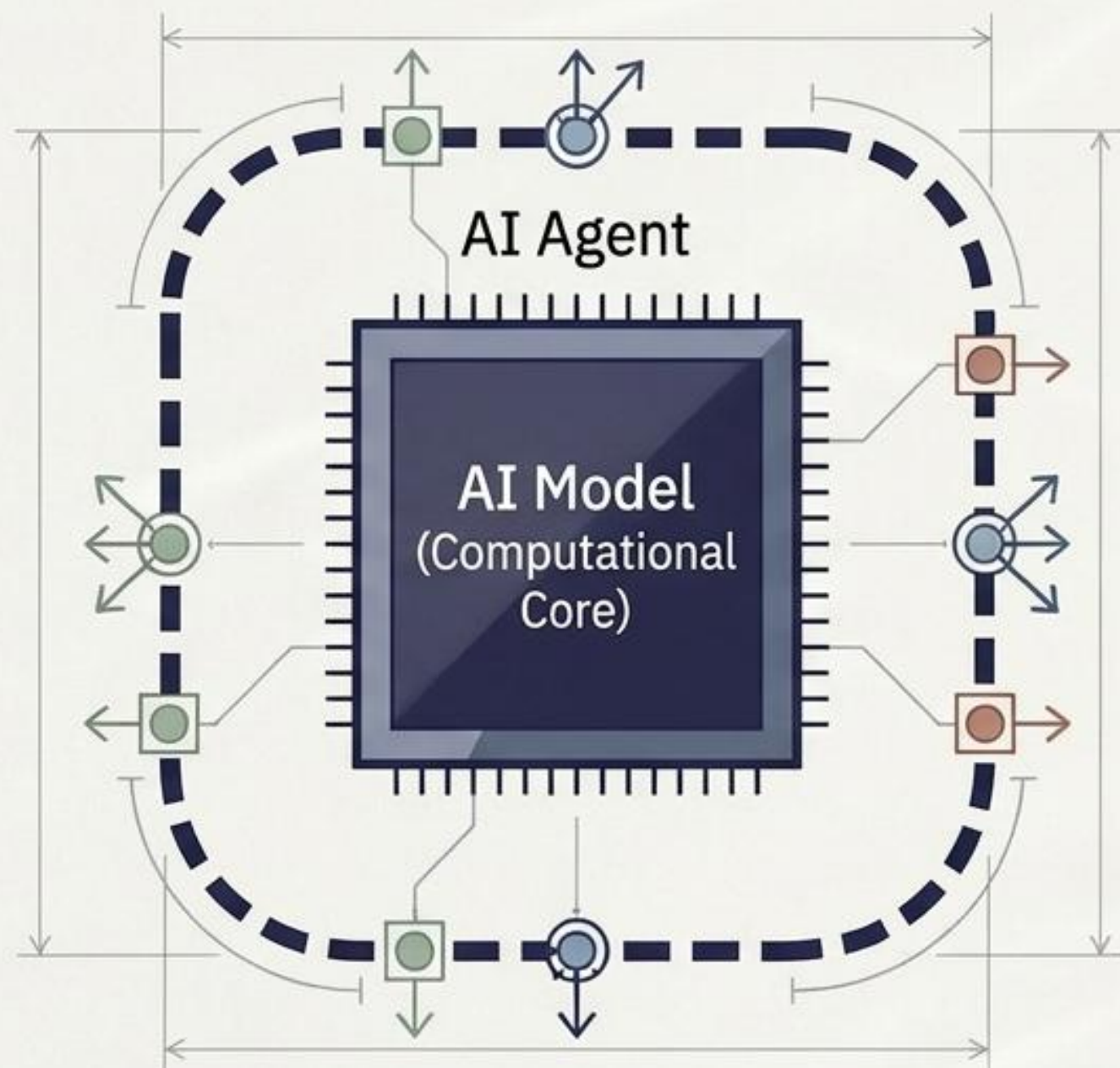
The Core Agent Model

An intelligent agent is a system that perceives its environment through sensors and acts upon that environment through actuators.

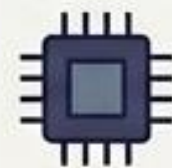


Environment → Sensors → Agent → Actuators → Environment

Expanding the Boundary: Agent $\neq$ Model



Agent vs. Model



GPT-like model = A computational model (the brain alone).



Chatbot application = A digital software agent.



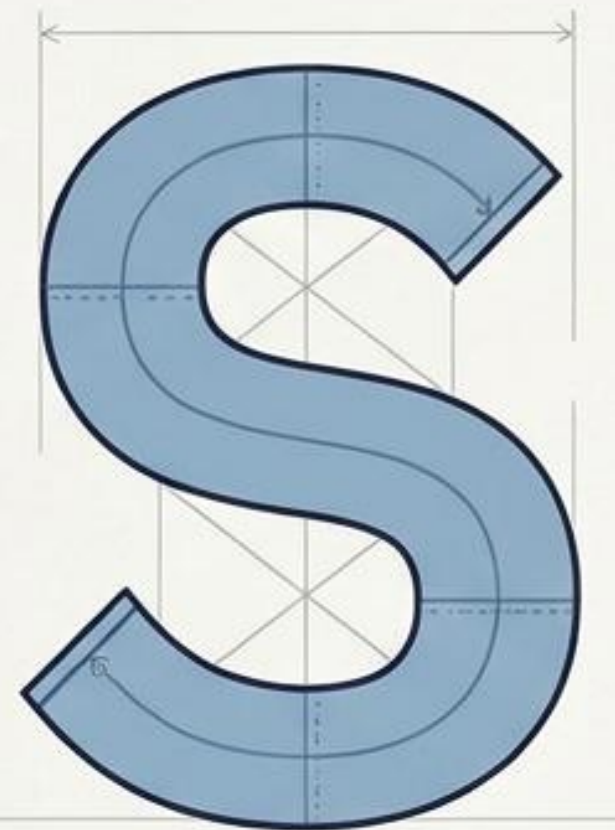
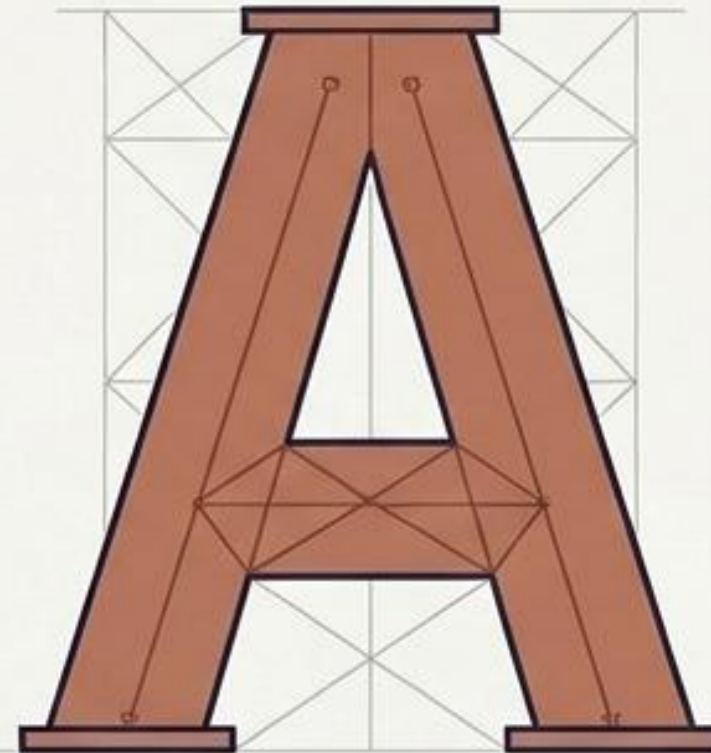
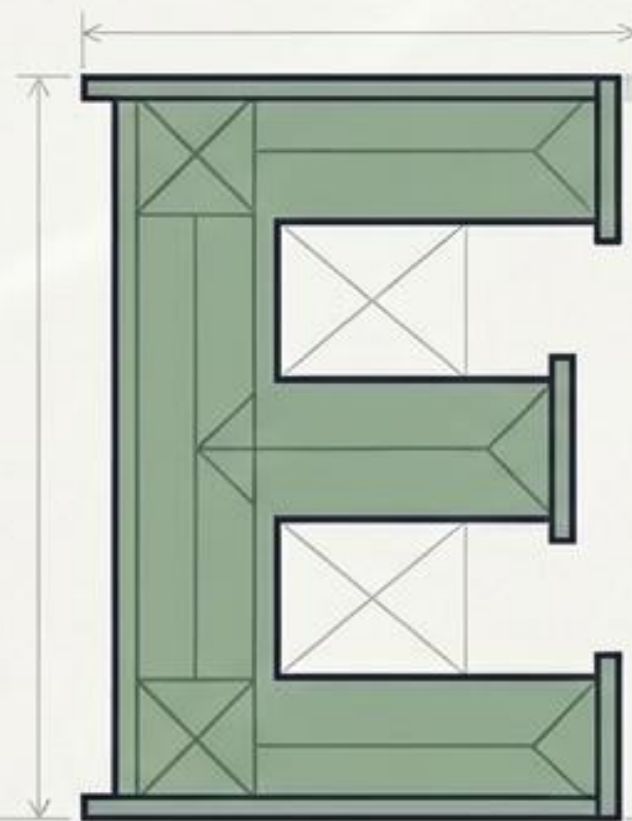
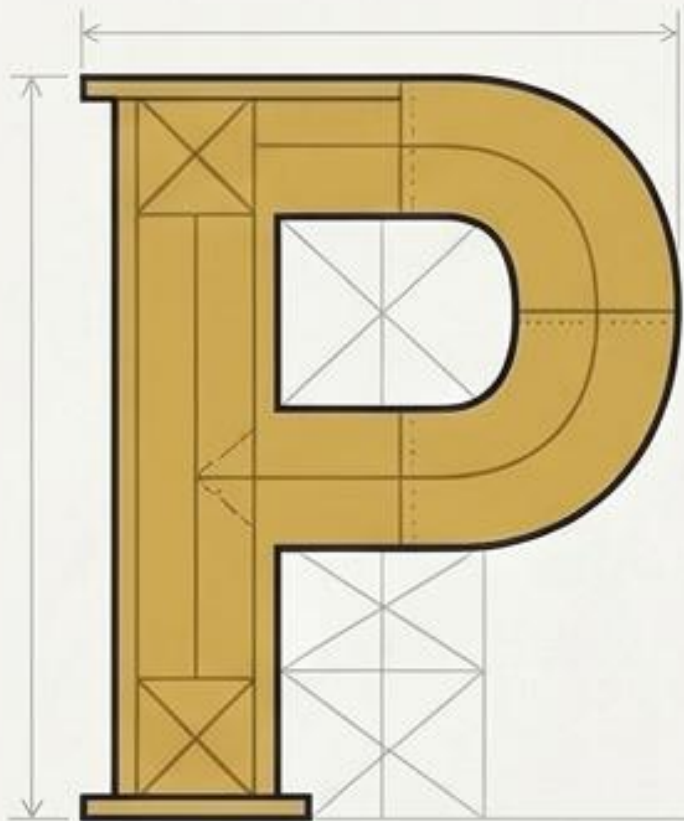
Autonomous robot = A physical agent.



An agent must always be considered in relation to its environment and its actions, not merely its internal computational logic.

Why Do We Need PEAS?

Saying 'Build an intelligent robot' is an incomplete requirement. PEAS converts a vague AI problem into a well-defined task specification.



Performance Measure

What is the goal?

Environment

Where does it operate?

Actuators

How can it act?

Sensors

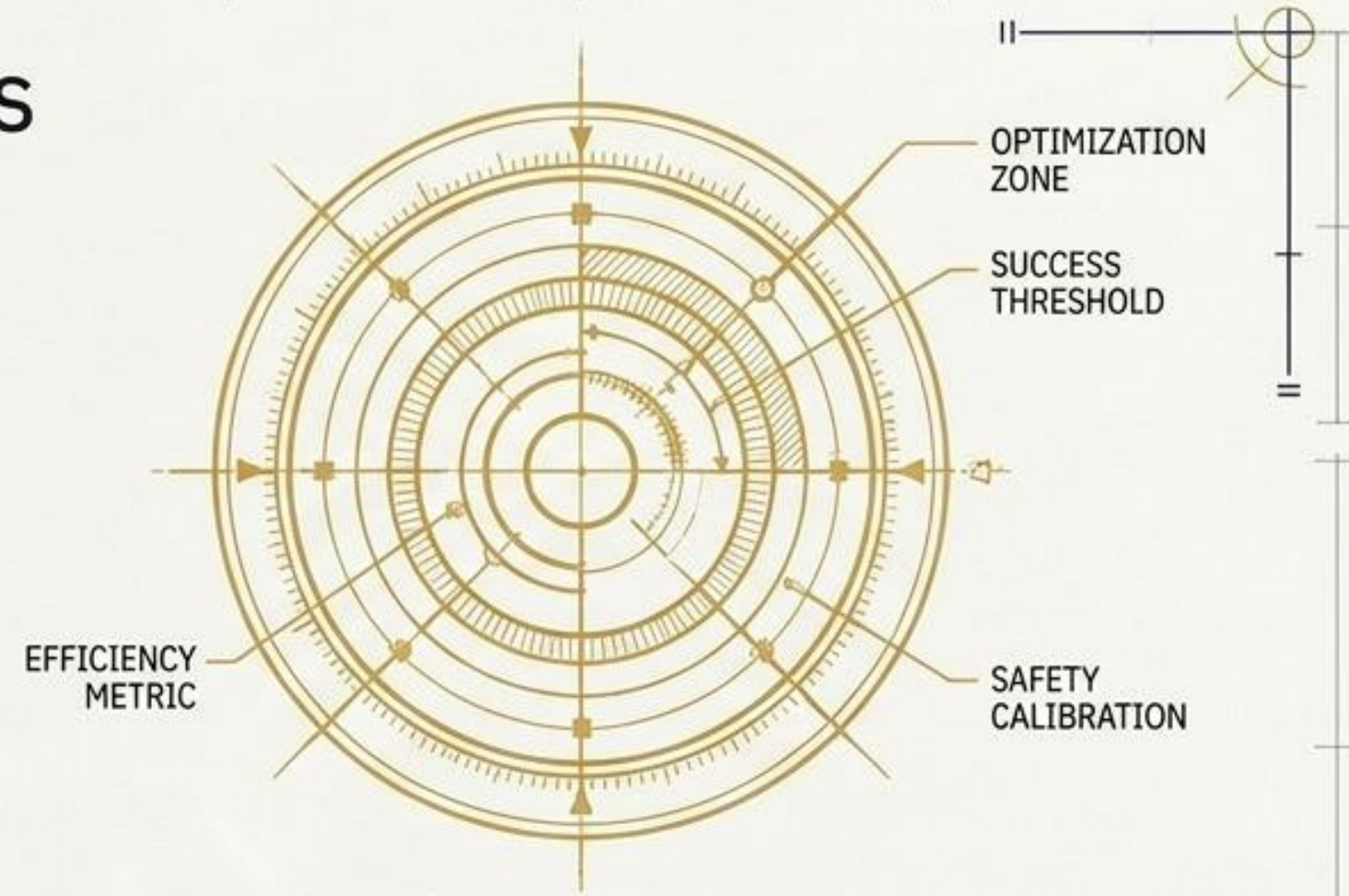
What can it observe?

Performance (P): Defining Success

The criteria used to evaluate the agent's behavior and the results obtained after processing. (It is not just the stated goal).

Taxi Examples:

- Passenger safety, travel time, efficiency, comfort, legal compliance.



The Optimization Trap.

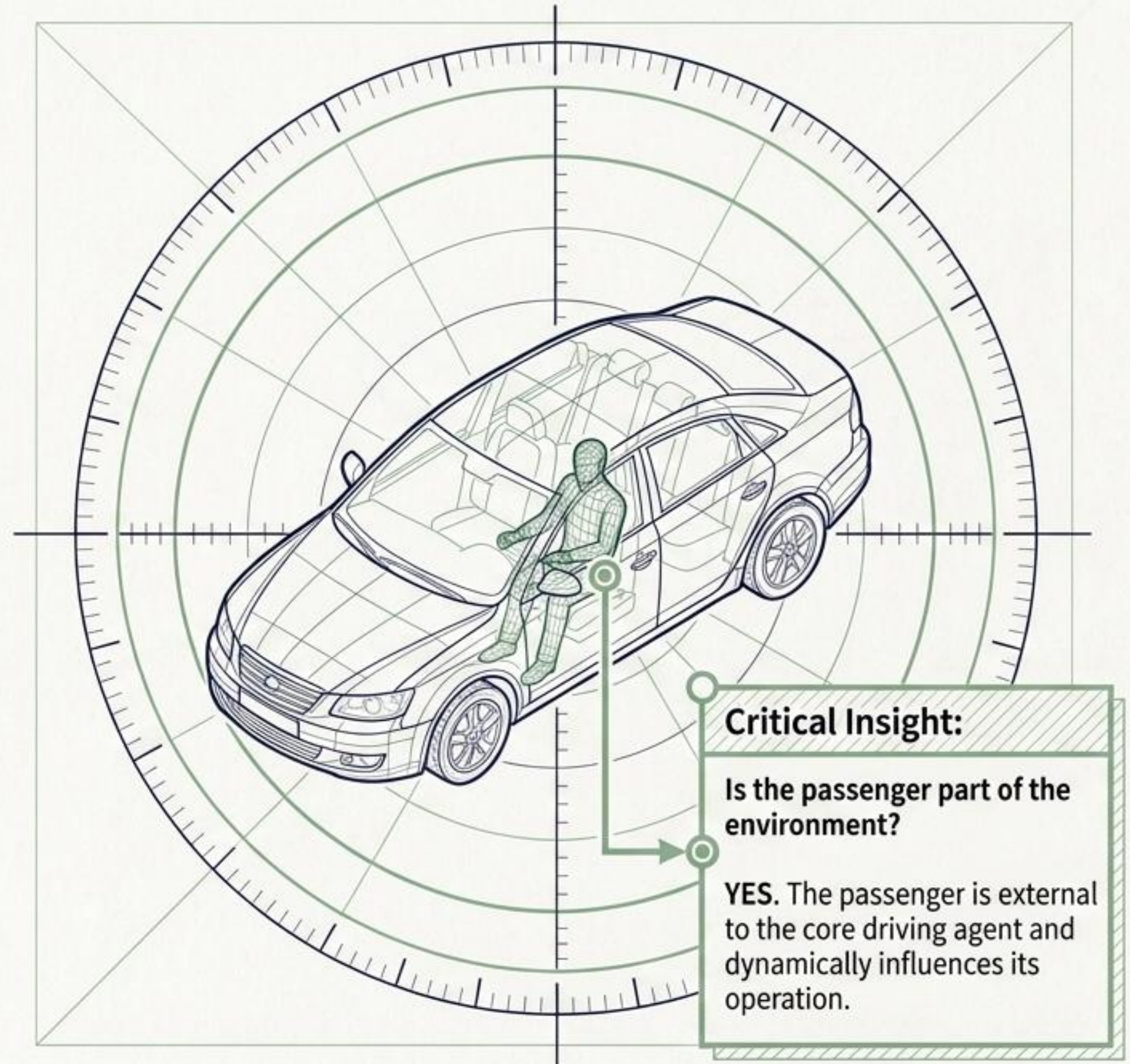
A poorly designed performance measure produces undesirable behavior. If an autonomous taxi only **optimizes for minimum travel time**, it will drive **aggressively and dangerously**. Good AI needs appropriate measures, not merely powerful algorithms.

Environment (E): The Operating Theatre

Everything external to the agent that affects its operation and with which it interacts.

Taxi Examples:

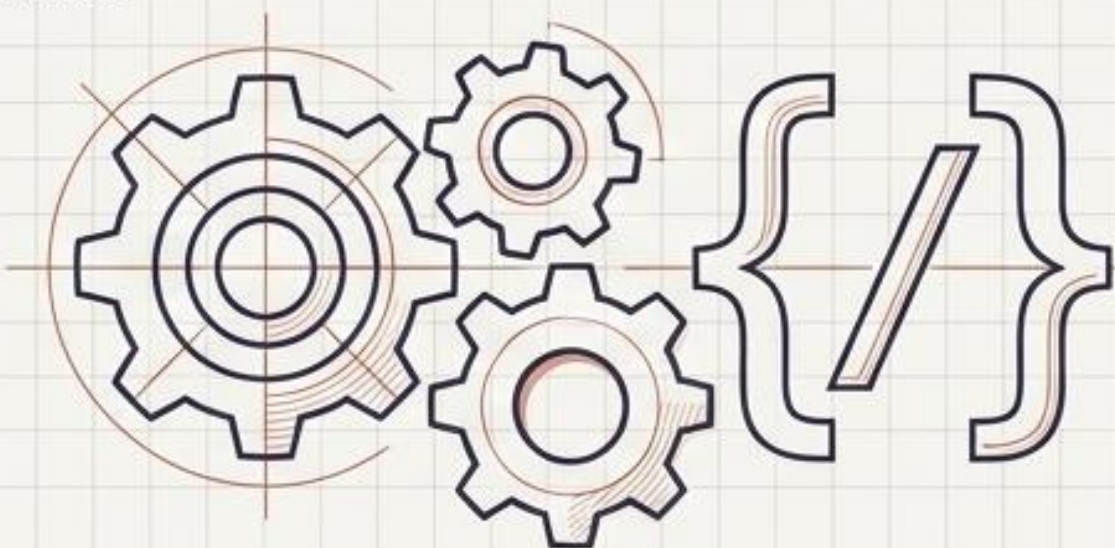
- Roads, Traffic, Weather
- Signals, Other vehicles
- Signals
- Signals
- Other vehicles
- Signs.



Action (A) & Perception (S)

Actuators

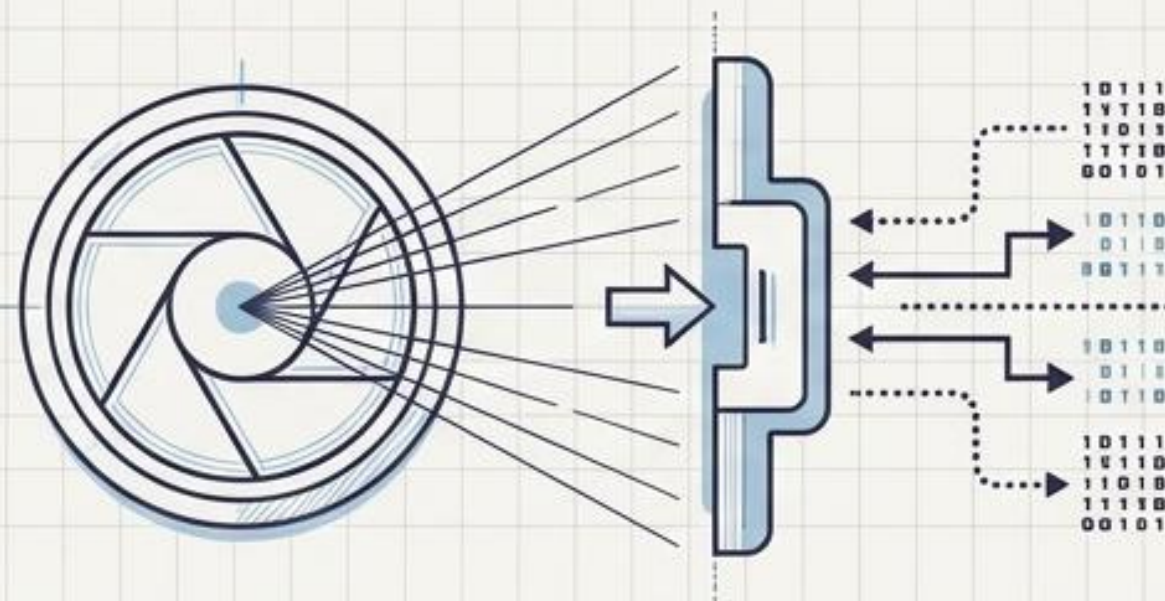
Devices/software through which agents perform actions.



- Physical: Wheels, robotic arms, steering.
- Software: API calls, database operations, sending messages.

Sensors

Devices from which agents perceive observations.



- Physical: Cameras, LiDAR, GPS.
- Software: User text, conversation history, external API data.

Takeaway: Sensors define the limits of knowledge. If an agent cannot sense something directly, it must infer it.

The Power of 'P': Task vs. Technology

Does the PEAS specification change if the hardware is identical? Compare an Autonomous Taxi to an Autonomous Ambulance.



Autonomous Taxi



Autonomous Ambulance

Performance (P)	Comfort, legal compliance, fuel efficiency.	Patient safety, rapid response, emergency traffic rules, medical urgency.
Environment (E)	Roads, traffic, pedestrians, weather	
Actuators (A)	Steering, accelerator, brake, indicators	
Sensors (S)	Cameras, GPS, LiDAR, radar	

Synthesis Insight: PEAS depends entirely on the specific task, not merely the underlying technology.

Agent Application Matrix

Agent Type	Sensors (Input)	Actuators (Output)
Physical Robot	Camera, LiDAR	Motors, Grippers
Self-Driving Car	Camera, GPS, Radar	Steering, Brakes
Chatbot	User text input	Text response
Game AI	Game state data	Game actions/moves

PEAS in Practice: Medical Diagnosis AI

Performance: Diagnostic accuracy, patient safety, low false-negative rate, appropriate referrals, timely recommendations.

Actuators: Generate diagnosis, recommend tests, request info, alert clinician.

AI Brain

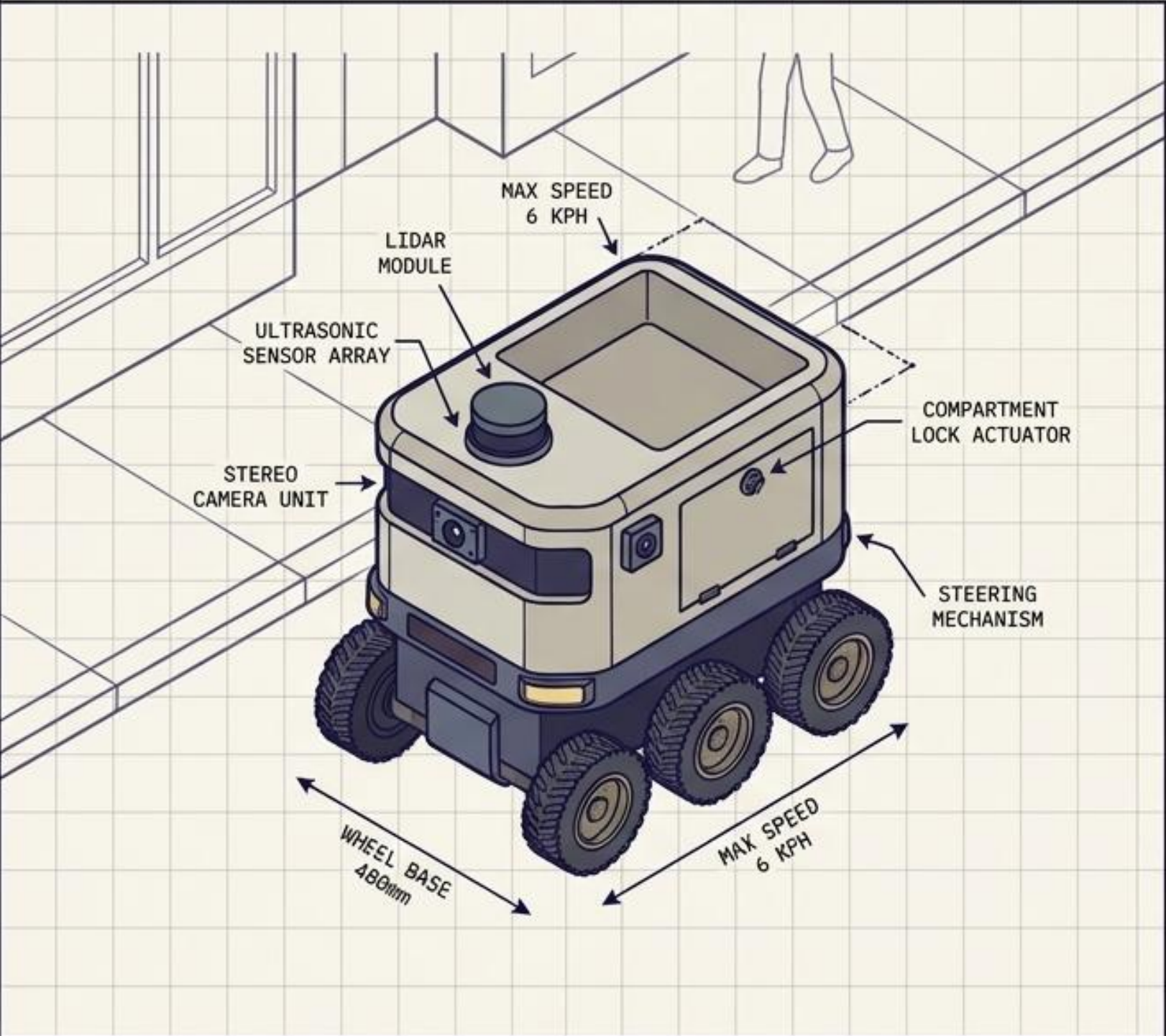
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graph TD; AI_Brain[AI Brain] --- Performance[Performance: Diagnostic accuracy, patient safety, low false-negative rate, appropriate referrals, timely recommendations.]; AI_Brain --- Actuators[Actuators: Generate diagnosis, recommend tests, request info, alert clinician.]; AI_Brain --- Environment[Environment: Hospital, patients, medical records, laboratory, clinical guidelines.]; AI_Brain --- Sensors[Sensors: Symptoms, medical history, lab results, imaging, vital signs.];
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Environment: Hospital, patients, medical records, laboratory, clinical guidelines.

Sensors: Symptoms, medical history, lab results, imaging, vital signs.

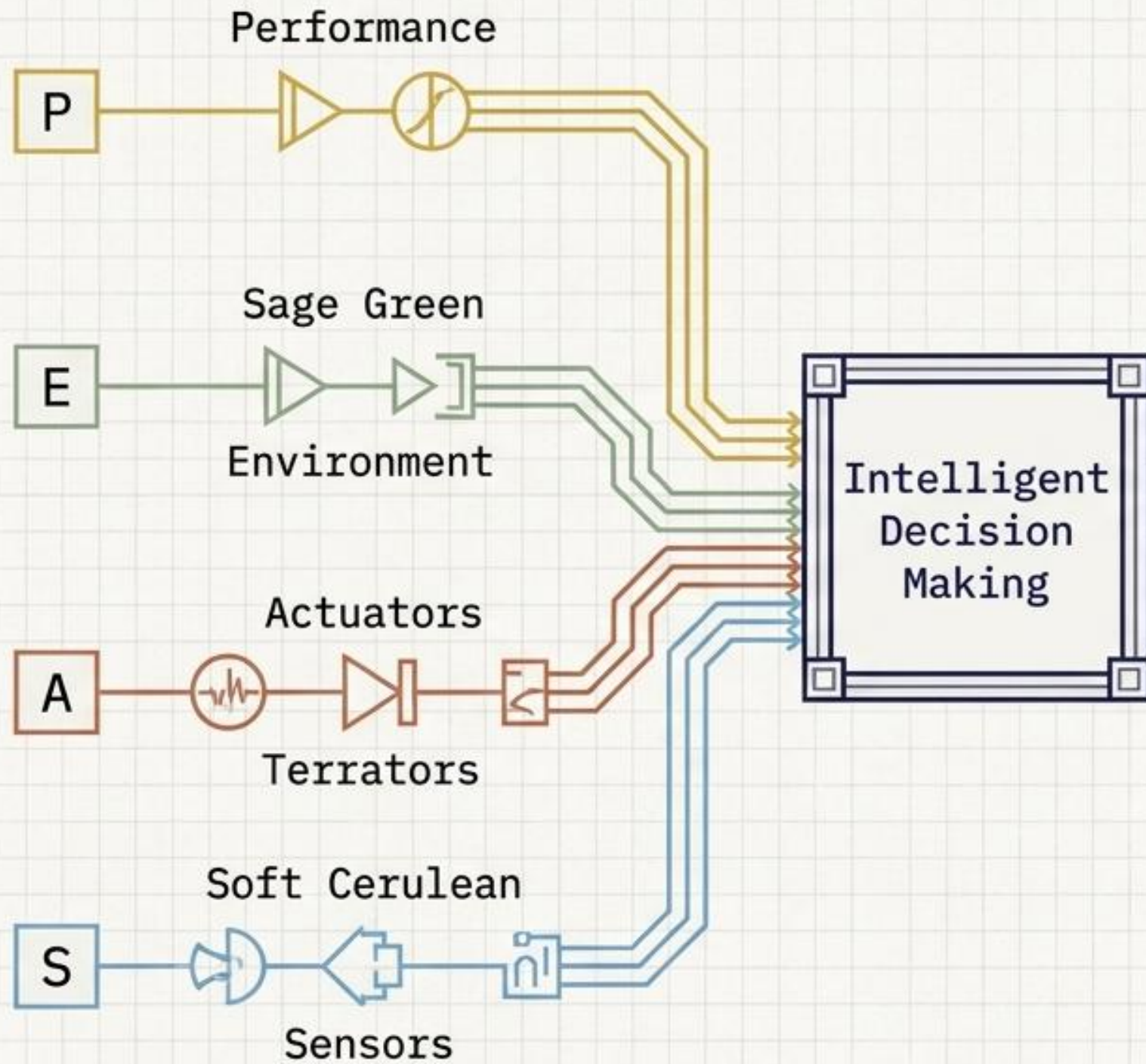
Formative Assessment: Food Delivery Robot

Design a PEAS description for an AI-powered delivery bot.



P:	Delivery accuracy, delivery time, safety, energy efficiency.
E:	Roads, sidewalks, pedestrians, buildings, weather.
A:	Wheels, steering, brakes, delivery compartment lock.
S:	Cameras, GPS, ultrasonic sensors, LiDAR.

The Foundation of AI Design



- Before writing a single line of code, an AI's operational reality must be mapped.
- PEAS translates vague requests into rigorous specifications, directly dictating the chosen agent architecture.
- Mastering the PEAS framework is the essential first step in moving from theoretical computational models to applied, real-world Artificial Intelligence.